Legendre Differential Equation and Legendre Polynomials

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Introduction

- The second-order linear ODE with variable coefficients

$$(1 - x^2)y'' - 2xy' + p(p+1)y = 0, \quad p \text{ a real number},$$

  is known as **Legendre's differential equation**, named after Adrien Marie Legendre (1752–1833).
- Legendre is best known for his work on elliptic integrals and the theory of numbers.
- Legendre's equation occurs in many physical and engineering problems involving **spherical geometry and gravitation** — e.g. potential theory, heat conduction, and wave propagation on a sphere.
- Solving it leads to a remarkable family of polynomials — the **Legendre polynomials** — they belong to Special Functions.
- The equation involves a parameter p, whose value depends on the physical problem at hand.



Legendre's Equation: Standard and Self-Adjoint Form

- Legendre's equation (or simply Legendre's equation) is

$$(1-x^2)y'' - 2xy' + p(p+1)y = 0. \tag{1}$$

- It can also be written in **self-adjoint form**:

$$\frac{d}{dx}\left\{(1-x^2)\frac{dy}{dx}\right\} + p(p+1)y = 0.$$

- The only singular points of (1) are $x = 1$, $x = -1$, and $x = \infty$ — and all three are **regular** singular points.
- Therefore Legendre's differential equation is a **Fuchsian differential equation** : (A Fuchsian differential equation is a linear ordinary differential equation in the complex domain whose singular points—including the point at infinity—are all regular singular points)
- So (1) is really a whole *family* of ODEs, one for each value of p.
- Any solution of (1) is called a **Legendre function**.



Legendre's Equation and Its Standard Form

- Consider the Legendre equation

$$(1-x^2)y'' - 2xy' + p(p+1)y = 0, \qquad |x| < 1.$$

- Here p is a constant, often called a parameter. The behavior of solutions differs for different values of p.
- In standard form,

$$y'' - \frac{2x}{1-x^2}y' + \frac{p(p+1)}{1-x^2}y = 0.$$

- $P(x) = -2x/(1-x^2)$ and $Q(x) = p(p+1)/(1-x^2)$ are both analytic at $x = 0$.
- So $x = 0$ is an **ordinary point** of the ODE, and it has a power series solution

$$y(x) = \sum_{n=0}^{\infty} a_n x^n.$$



Series Solution: The Recurrence Relation

- Substitute $y(x) = \sum_{n=0}^{\infty} a_n x^n$ into the ODE and set the coefficient of x^n to zero:

$$(1-x^2)\sum n(n-1)a_n x^{n-2} - 2x\sum n a_n x^{n-1} + p(p+1)\sum a_n x^n = 0$$

$$\Rightarrow\ (n+2)(n+1)a_{n+2} - n(n-1)a_n - 2na_n + p(p+1)a_n = 0$$

$$\Rightarrow\ a_{n+2} = -\frac{(p-n)(p+n+1)}{(n+1)(n+2)}\,a_n. \qquad (2)$$

- Leaving a_0, a_1 arbitrary, we compute the rest. For instance:

$$a_2 = -\frac{p(p+1)}{1\cdot 2}a_0, \qquad a_3 = -\frac{(p-1)(p+2)}{2\cdot 3}a_1,$$

$$a_4 = \frac{p(p-2)(p+1)(p+3)}{4!}a_0, \qquad a_5 = \frac{(p-1)(p-3)(p+2)(p+4)}{5!}a_1.$$



The General Solution: $y_1(x)$ and $y_2(x)$

- The solution is $y(x) = a_0 y_1(x) + a_1 y_2(x)$, where

$$y_1(x) = 1 - \frac{p(p+1)}{2!}x^2 + \frac{p(p-2)(p+1)(p+3)}{4!}x^4 - \cdots$$

$$y_2(x) = x - \frac{(p-1)(p+2)}{3!}x^3 + \frac{(p-1)(p-3)(p+2)(p+4)}{5!}x^5 - \cdots$$

- $y_1(x)$ contains only even powers of x; $y_2(x)$ contains only odd powers.
- Together, y_1 and y_2 form a fundamental system of solutions of Legendre's equation.
- For the series of $y_1(x)$, the ratio of consecutive terms is

$$\left|\frac{a_{2n+2}x^{2n+2}}{a_{2n}x^{2n}}\right| = \left|\frac{(p-2n)(p+2n+1)}{(2n+1)(2n+2)}\right| \to |x|^2 \quad \text{as } n \to \infty.$$



Radius of Convergence

- So the radius of convergence of the series for $y_1(x)$ is 1.
- The same holds for $y_2(x)$.
- Thus the series solutions are valid for $-1 < x < 1$; at $x = \pm 1$, $P(x)$ and $Q(x)$ fail to be analytic, so nothing is guaranteed there.
- We assumed p is a non-negative integer from the outset. But *why* does this make one of the two series solutions a polynomial?
- One can observe the presence of $p - n, n \in \mathbb{N}_0$ in the series y_1 and y_2, where $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.
- Now exactly one of y_1, y_2 terminates into a polynomial precisely due even or odd nature of $n \in \mathbb{N}_0$, when p is a non-negative integer.



Behavior at Integer p: Three Cases

- When $p \notin \mathbb{Z}$, the numerators in the coefficients never vanish, and both $y_1(x)$ and $y_2(x)$ remain genuine infinite series.
- When $p \in \mathbb{Z}$, something special happens — we consider three cases separately:
  - **Case 1:** $p = 0$.
  - **Case 2:** p nonzero and even.
  - **Case 3:** p odd.
- In each case, exactly one of $y_1(x)$ or $y_2(x)$ turns out to be a **polynomial** — the other remains an infinite series.



Case 1: $p = 0$

- If $p = 0$, then $y_1(x) = 1$, which is a constant.
- And

$$y_2(x) = x - \frac{(-1)(2)}{3!}x^3 + \frac{(-1)(-3)(2)(4)}{5!}x^5 - \cdots$$

  remains a genuine power series.
- Here $y_1(x)$ is a (trivial) polynomial, and $y_2(x)$ is an infinite series.



Case 2: p Nonzero and Even

- Suppose $p = 2k$ for some integer $k \geq 1$. Then

$$y_1(x) = 1 - \frac{2k(2k+1)}{2!}x^2 + \cdots$$

$$+ (-1)^k \frac{2k(2k-2)\cdots(2)(2k+1)(2k+3)\cdots(4k-1)}{(2k)!}x^{2k}.$$

- The *next* term in the series has a numerator factor $(p - 2k) = 0$.
- All succeeding terms vanish, so $y_1(x)$ **terminates** — it is a polynomial of degree $2k = p$.
- Meanwhile $y_2(x)$ remains an infinite series.



Case 3: p Odd

- Suppose $p = 2k + 1$ for some integer $k \geq 0$. Then

$$y_2(x) = x - \frac{2k(2k+3)}{3!}x^3 + \cdots$$

$$+ (-1)^k \frac{2k(2k-2)\cdots(2)(2k+3)(2k+5)\cdots(4k+1)}{(2k+1)!}x^{2k+1}.$$

- The next term has numerator factor $(p - (2k + 1)) = 0$, so all succeeding terms vanish.
- $y_2(x)$ **terminates** — it is a polynomial of degree $2k + 1 = p$.
- Meanwhile $y_1(x)$ remains an infinite series.



Polynomial Solutions: A Summary

- So if $p \in \mathbb{Z}$, exactly one of $y_1(x)$ or $y_2(x)$ is a polynomial.
- $p = 0$: ODE is $(1 - x^2)y'' - 2xy' = 0$; polynomial solution $y_1(x) = 1$, with $y_1(1) = 1$.
- $p = 2$: ODE is $(1 - x^2)y'' - 2xy' + 6y = 0$; polynomial solution (with $p = 2k,\ k = 1$)

$$y_1(x) = 1 - \frac{2(3)}{2!}x^2 = 1 - 3x^2, \qquad y_1(1) = -2.$$

- $p = 1$: ODE is $(1 - x^2)y'' - 2xy' + 2y = 0$; polynomial solution $y_2(x) = x$, with $y_2(1) = 1$.
- $p = 3$: ODE is $(1 - x^2)y'' - 2xy' + 12y = 0$; polynomial solution (with $p = 2k + 1,\ k = 1$)

$$y_2(x) = x - \frac{2(5)}{3!}x^3 = x - \frac{5}{3}x^3, \qquad y_2(1) = -\frac{2}{3}.$$



Normalizing the Polynomial Solutions

- The polynomial solutions, evaluated at $x = 1$:

Parameter p	0	1	2	3
Degree of polynomial	0	1	2	3
Which solution	y_1	y_2	y_1	y_2
Its value at 1	1	1	-2	$-2/3$

- If $y_1(x)$ solves a Legendre equation, so does any constant multiple of $y_1(x)$ — the same holds for $y_2(x)$.
- So we may rescale each polynomial by a suitable constant so that its value at $x = 1$ becomes exactly 1.



Definition of the Legendre Polynomial

- The **Legendre polynomial of degree** n, denoted $P_n(x)$, is the polynomial of degree n satisfying Legendre's equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0, \qquad y(1) = 1.$$

- If n is even, $P_n(x) = y_1(x)$ (rescaled) and contains no odd powers of x.
- If n is odd, $P_n(x) = y_2(x)$ (rescaled) and contains no even powers of x.
- By construction, $P_n(1) = 1$ for every n.



First Few Legendre Polynomials

- Using the earlier computations:

$$P_0(x) = y_1(x)\ (p = 0) = 1.$$

$$P_1(x) = y_2(x)\ (p = 1) = x.$$

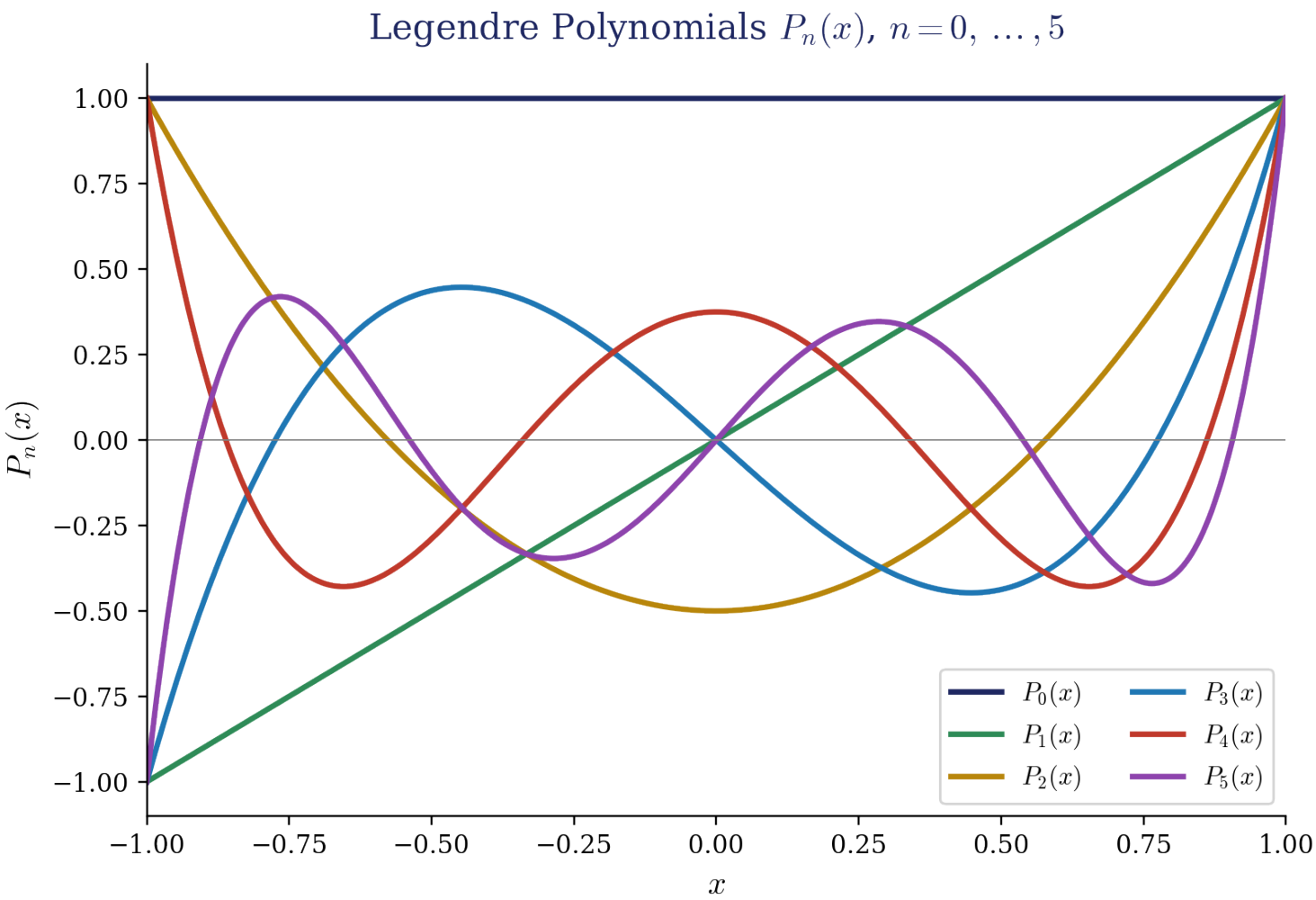
$$P_2(x) = y_1(x)\ (p = 2) = \frac{1}{-2}(1 - 3x^2) = \frac{1}{2}(3x^2 - 1).$$

$$P_3(x) = y_2(x)\ (p = 3) = -\frac{3}{2}\left(x - \frac{5}{3}x^3\right) = \frac{1}{2}(5x^3 - 3x).$$

- These match the constants needed to make $P_n(1) = 1$.



Graphs of $P_n(x)$



Graphs of Legendre Polynomials

- Plots of $P_0(x), \ldots, P_5(x)$ on $[-1, 1]$:
  - All curves pass through the point $(1, 1)$ — consistent with $P_n(1) = 1$.
  - $P_n(x)$ has exactly n real zeros, all lying in $(-1, 1)$.
  - Even-degree polynomials are symmetric about the y-axis; odd-degree ones are antisymmetric.
  - As n increases, the oscillations become more rapid but stay bounded within $[-1, 1]$.
  - P_n is even in x for even n, odd for odd n.
- This bounded oscillatory behavior is part of why Legendre polynomials are so useful for expanding functions on $[-1, 1]$.



Choosing the Leading Coefficient Directly

- Rather than rescaling after the fact, we can build in the condition $P_n(1) = 1$ from the start.
- Let a_n be the coefficient of x^n in $P_n(x)$. Choose

$$a_n = \frac{(2n)!}{2^n (n!)^2} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!}, \qquad n \geq 0.$$

- The recurrence relation (2) for the coefficients then gives (working backwards from a_n - Replace n by $n-2$ and p by n in (2))

$$a_{n-2} = -\frac{n(n-1)}{2(2n-1)} a_n = -\frac{(2n-2)!}{2^n (n-1)!(n-2)!}.$$

$$a_{n-2k} = (-1)^k \frac{(2n-2k)!}{2^n \, k! \, (n-k)! \, (n-2k)!}, \qquad n \geq 2k.$$

